

Shubhang Bhatnagar

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EDUCATION

University of Illinois Urbana-Champaign

(Aug '21 - Dec '26 expected)

Ph.D. student in Electrical and Computer Engineering, Advisor: [Prof. Narendra Ahuja](#)

Research: Understanding representations in multimodal LLMs to improve their post-training and their inference efficiency

Indian Institute of Technology, Bombay

(Jul '16 - Jul '21)

Dual Degree (B.Tech + M.Tech) Electrical Engineering, Specializing in Signal Processing, Minor in CS

Masters Thesis: Active Learning Advisor: [Prof. Amit Sethi](#)

GPA 9.81/10, Institute Silver Medal, Department Rank 1/72

Nanyang Technological University, Singapore

(Jul '19 - Dec '19)

GPA 4.82/5, Temasek Funded Semester Exchange Program in Electrical Engineering

PROFESSIONAL EXPERIENCE

Meta Reality Labs, WA | Research Scientist Intern | Adv: [Dr. Shane Moon](#)

(May '25 - Aug '25)

Project: VideoMind: Thinking in Steps for Long Video Understanding [[paper](#)] *EACL '26 (Industry, Oral)*

- Proposed **VideoMind**, a novel agentic framework that reframes long video understanding as a self-reasoning loop, where the **MLLM interrogates the video** via recursive query decomposition and multi-scale temporal and spatial analysis; VideoMind establishes a **new state-of-the-art performance on Video-MME** (77.6%, using Qwen 2.5VL +4.8% gain), MLVU, and LongVideoBench, significantly outperforming fixed-context baselines. Evaluated at scale via a parallel vLLM pipeline on 16x H100s.

HP Inc, CA | ML Research Intern | Adv: [Dr. Kar-Han Tan](#)

(May '24 - May '25)

Project: LUQ: Layerwise Ultra-Low Bit Quantization for MLLMs [[webpage](#)] *TMLR '26, Patent Under Review*

- Developed LUQ, a novel framework to apply ultra-low-bit (< 4 bit) quantization to Multimodal LLMs. LUQ selectively quantizes layers that are most resilient to quantization, measured by their output activation entropy which we show varies significantly across layers in multimodal LLMs. LUQ achieves SOTA memory-compression trade-off on 9 different MLLM benchmarks, e.g. using 31% less memory vs. 4-bit while maintaining MME performance within 2% of the 4-bit model.
- Deployed LUQ on prototype systems ([Demo Link](#)) demonstrating 50% inference speedups on laptops and more than 10x on edge devices. **LUQ was demonstrated to HP's CEO for its innovation and potential applications.**
- Additional Project:** Led the development and prototyping for a novel redacted speculative decoding algorithm, speeding up LLM inference on laptops by up to 3x by selectively utilizing cloud GPUs for non-private information. **US Patent** under review.

Bosch Research, CA | Comp. Vision & Mixed Reality Intern | Adv: [Dr. Sharath Gopal](#)

(May '22 - Aug '22)

Project: Long-Distance Gesture Recognition using Dynamic Neural Networks [[webpage](#)], *IROS '23 (Oral)*

- Proposed a **novel dynamic neural network** to enable gesture recognition from long distances by identifying and processing only gesturing subject features. Outperformed **SOTA** in terms of recognition accuracy while using **40% lesser compute**.
- Developed a demo of the system to control mobile robots. **US Patents** granted ([link](#)) and under review (18/068,118).

Qualcomm | Modem Firmware Intern

(May '19 - Jul '19)

Decimal Point Analytics | Computer Vision Intern

(Jun '18 - Aug '18)

RESEARCH HIGHLIGHTS

Full list at: <https://shubhangb97.github.io>

Beyond Scalar Distances: Attribute Gradients from Frozen MLLMs, [[project page](#)], *Under Review*

- Proposed **SAGA**, a framework to train **vision encoders** with **attribute-resolved gradients from a frozen MLLM** instead of class labels, which supervise all embedding dimensions (distance) in unison. SAGA uses **GRPO** on the MLLM's verdicts to route gradients only on **discriminative directions**. Cut zero-shot retrieval error by **27-52%** over the prior **SOTA** on benchmarks.

Potential Field based Metric Learning, [[project page](#)], *CVPR'25*

Prof. Narendra Ahuja, UIUC

- Proposed a continuous potential-field representation to model the influence of embeddings as opposed to discrete n-tuplet/contrastive models used in embedding learning. **Proved that** such a potential field forms a better model of the data distribution than previous methods, helping learn better **representations for visual search** while being more robust to **label noise**. Significantly reduced error on zero-shot image retrieval benchmarks by **more than 10%** compared to **SOTA**.

Piecewise-Linear Manifolds for Deep Metric Learning [\[project page\]](#), *PMLR & CPAL '24 (Oral)*

UIUC

- Proposed an **unsupervised embedding learning** framework to learn a semantically meaningful embedding space for an unlabeled dataset. By using a piecewise linear model to model the neural data manifold, we identify images with similar content with **50% better accuracy** than current SOTA methods. Method established a new **SOTA** on unsupervised zero-shot retrieval benchmarks.

PAL - Pretext Based Active Learning [\[paper\]](#)[\[slides\]](#), *BMVC '21*

Masters Thesis, IITB

- Proposed a **technique** to estimate the novelty of a sample for a given model by using self-supervision to augment task-specific uncertainty leading to a better and more diverse selection of samples for active learning. Required 15% less labeled data than **state-of-the-art** techniques to achieve the same performance on classification and semantic segmentation.

PUBLICATIONS

- Beyond Scalar Distances: Semantic Attribute Gradients from Frozen MLLMs** [\[paper\]](#)[\[project page\]](#), S. Bhatnagar*, D. Baiju*, Narendra Ahuja, *Under Review*
- VideoMind: Thinking in Steps for Long Video Understanding**, [\[paper\]](#) **EACL'26** (Industry track) S. Bhatnagar, R. Wang, K. Krishnakumar, A. Ahmadyan, Z. Lin, L. Mathias, L. Dong, B. Damavandi, N. Ahuja, S. Moon
- LUQ: Layerwise Ultra-Low Bit Quantization for MLLMs** [\[paper\]](#)[\[project page\]](#), S. Bhatnagar*, A. Xu*, K.H. Tan, Narendra Ahuja, in *TMLR 2026*
- Potential Field Based Deep Metric Learning** [\[paper\]](#)[\[project page\]](#), S. Bhatnagar, Narendra Ahuja, in *CVPR 2025*
- System for Learning Long-Distance Recognition and Personalization of Gestures** [\[US Patent# US12387472B2\]](#)
- Rethinking Prompting Strategies for Multi-Label Recognition with Partial Annotations** [\[paper\]](#)[\[project page\]](#), S.Rawlekar, S. Bhatnagar, Narendra Ahuja, **WACV 2024**
- Improving Multi-label Recognition using Class Co-Occurrence Probabilities** [\[paper\]](#)[\[project page\]](#), S. Bhatnagar*, S.Rawlekar*, Narendra Ahuja, in **CVPRw 2024, ICPR 2024 (Oral Top-5%)**
- Piecewise-Linear Manifolds for Deep Metric Learning** [\[paper\]](#)[\[project page\]](#), S. Bhatnagar, Narendra Ahuja, in **PMLR Vol 234, CPAL 2024 (Oral)**
- Long-Distance Gesture Recognition using Dynamic Neural Networks** [\[paper\]](#)[\[project page\]](#), S. Bhatnagar, S. Gopal, N. Ahuja, L. Ren, in **IROS 2023 (Oral)**
- Insights on coding gain and its properties for principal component filter banks** [\[paper\]](#), P. Chaphekar, A. Bhatia, S. Bhatnagar, et al., Sādhanā, Journal of the Indian Academy of Sciences, 2023
- PAL - Pretext Based Active Learning** [\[paper\]](#), S. Bhatnagar, S. Goyal*, D. Tank*, A. Sethi, in **BMVC 2021**
- Analyzing Cross Validation in Compressed Sensing with Gaussian and Impulse Measurement Noise with L1 Errors** [\[paper\]](#), S. Bhatnagar*, C. Gurjarpadhye*, A. Rajwade, in **EUSIPCO 2021**
- QR Code Denoising Using Parallel Hopfield Networks** [\[preprint\]](#), S. Bhatnagar*, I. Bhatnagar*, Arxiv Preprint, 2018

*denotes that these authors contributed equally

HONORS AND AWARDS

- Institute Silver Medal**, IIT B for graduating at the top of my batch ('21)
- Bhavesh Gandhi memorial award** given to most outstanding student in EE IITB ('21)
- Institute Award for academic excellence**, IIT B (twice) ('18,'20)
- Temasek Foundation TFLearn fellowship** ('19)
- INSPIRE scholarship**, Govt of India for being in top 1% of class 12 ('15)
- KVPY fellowship** awarded by Govt of India with All India Rank 93 ('13)

TECH SKILLS

Python, PyTorch, vLLM, FSDP, HF Transformers, Diffusers, Llama cpp, Pytorch Lightning, Java, C, C++, ROS

SERVICE

- Reviewer** for ICPR '20; CVPR '22-26; ICCV'23-25, NAACL '24; NeurIPS '24-26; ICLR '24-25; Pattern Recognition '25
- Graduate Teaching Assistant for CS 444 Deep Learning for Computer Vision @ UIUC ('25)
- Teaching Assistant @ Math & EE IIT Bombay ('18,'20,'21)
- Institute Student Mentor @ SMP, IIT Bombay ('20)